Complex Numbers and their geometric representation:

A complex number z is an ordered pair (x, y) of real numbers x and y, written $z = (x, y)$ or $z = x + iy$ where $i = \sqrt{-1}$ is known as the imaginary unit.

Here, x is called the real part of z and y the imaginary part of z, written

$$x = \operatorname{Re} z \quad , \quad y = \operatorname{Im} z .$$

- If $x = 0$, $y \neq 0$, then $z = 0 + iy = iy$ which is purely imaginary.
- If $x \neq 0$, $y = 0$, then $z = x + i \cdot 0 = x$, which is real.

Hence, z is purely imaginary, if its real part is zero and is real, if its imaginary part is zero. This shows that every real number can be written in the form of a complex number by taking its imaginary part as zero. Hence, the set of real numbers is contained in the set of complex numbers.

- The even power of i is either 1 or -1 and odd power of i is either i or $-i$.

$i^2 = i \cdot i = \sqrt{-1} \cdot \sqrt{-1} = \sqrt{(-1)^2} = -1$

$i^3 = i^2 \cdot i = -1 \cdot i = -i$

$i^4 = (i^2)^2 = (-1)^2 = 1$

$i^5 = i^4 \cdot i = 1 \cdot i = i$, etc.

- Two complex numbers are equal if and only if their corresponding real and imaginary parts are equal.

If $z_1 = (x_1, y_1) = x_1 + iy_1$, $z_2 = (x_2, y_2)$, then $z_1 = z_2$ iff $x_1 = x_2$ and $y_1 = y_2$.

Algebra of complex numbers:

Let $z_1 = \overset{(x_1, y_1)}{x_1 + iy_1}$ and $z_2 = \overset{(x_2, y_2)}{x_2 + iy_2}$ be two complex numbers.

a) Addition:

$$z_1 + z_2 = (x_1 + iy_1) + (x_2 + iy_2)$$
$$= x_1 + x_2 + i(y_1 + y_2)$$

b) Subtraction:

$$z_1 - z_2 = (x_1 + iy_1) - (x_2 + iy_2)$$
$$= (x_1 - x_2) + i(y_1 - y_2)$$

c) Multiplication:

$$z_1 z_2 = (x_1 + iy_1) \cdot (x_2 + iy_2)$$
$$= x_1 x_2 + i x_1 y_2 + i y_1 x_2 - y_1 y_2 \qquad [\because i^2 = -1]$$
$$= (x_1 x_2 - y_1 y_2) + i(x_1 y_2 - x_2 y_1)$$

d) Division:

If $z_2 \neq 0$,

$$\frac{z_1}{z_2} = \frac{x_1 + iy_1}{x_2 + iy_2} = \left(\frac{x_1 + iy_1}{x_2 + iy_2}\right)\left(\frac{x_2 - iy_2}{x_2 - iy_2}\right)$$
$$= \left[\frac{x_1 x_2 + y_1 y_2}{x_2^2 + y_2^2}\right] + i\left[\frac{x_2 y_1 - x_1 y_2}{x_2^2 + y_2^2}\right]$$

Note :

Subtraction and division are defined as the inverse operation of addition and multiplication, respectively. Thus, the difference $z = z_1 - z_2$ is the complex number z for which $z_1 = z + z_2$.

Hence, $z_1 - z_2 = (x_1 - x_2) + i(y_1 - y_2)$.

The quotient $z = \frac{z_1}{z_2}$ $(z_2 \neq 0)$ is the complex number z for which $z_1 = z z_2$. If we equate the real and imaginary parts on both sides of this equation, setting $z = x + iy$, we obtain

$$x_1 = x_2 x - y_2 y, \quad y_1 = y_2 x + x_2 y.$$

The solution is

$$z = \frac{z_1}{z_2} = x + iy.$$

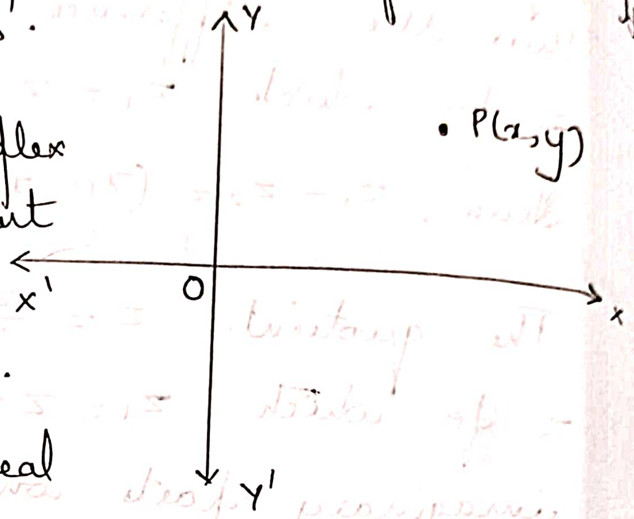
$$x = \frac{x_1 x_2 + y_1 y_2}{x_1^2 + y_2^2}, \quad y = \frac{x_2 y_1 - x_1 y_2}{x_2^2 + y_2^2}.$$

The practical rule used to get this is by multiplying numerator and denominator of $\frac{z_1}{z_2}$ by $x_2 - iy_2$ and satisfying:

$$z = \frac{x_1 + iy_1}{x_2 + iy_2} = \frac{(x_1 + iy_1)(x_2 - iy_2)}{(x_2 + iy_2)(x_2 - iy_2)} = \frac{x_1 x_2 + y_1 y_2}{x_2^2 + y_2^2} + i\,\frac{x_2 y_1 - x_1 y_2}{x_2^2 + y_2^2}$$

Geometrical representation of complex numbers:

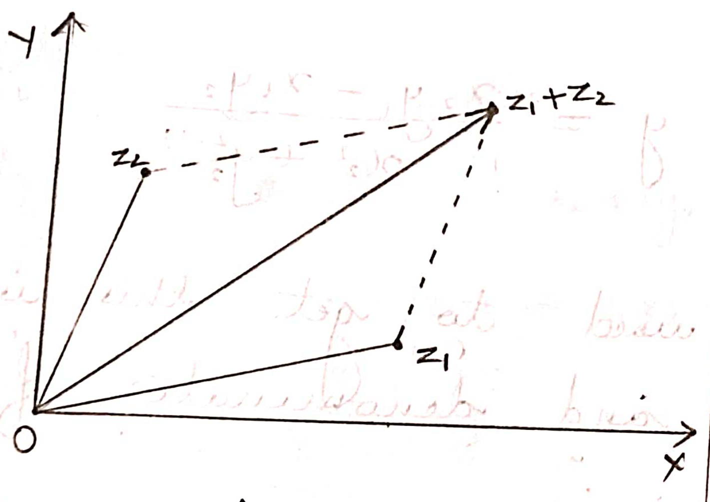
Any complex number $z = x + iy$ can be represented as point $P(x, y)$ in the xy-plane with reference to the rectangular x and y axes.



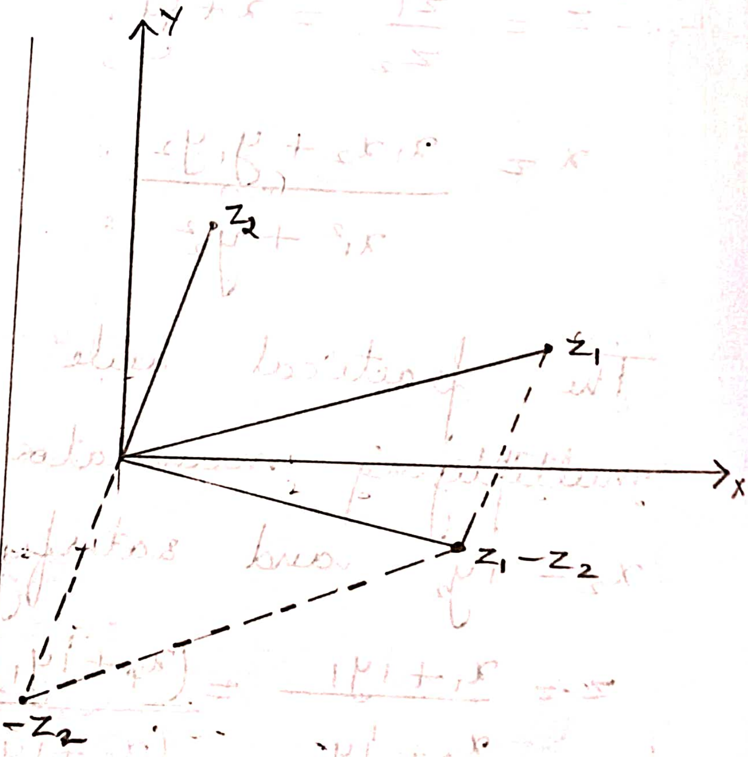
The plot of a given complex number $z = x + iy$, as the point $P(x, y)$ in the XY-plane is known as Argand's diagram.

The X-axis is called the real axis, Y-axis is called the imaginary axis and the XY-plane is called the complex plane.

Addition and subtraction can now be visualized as illustrated below:



Addition of complex numbers.

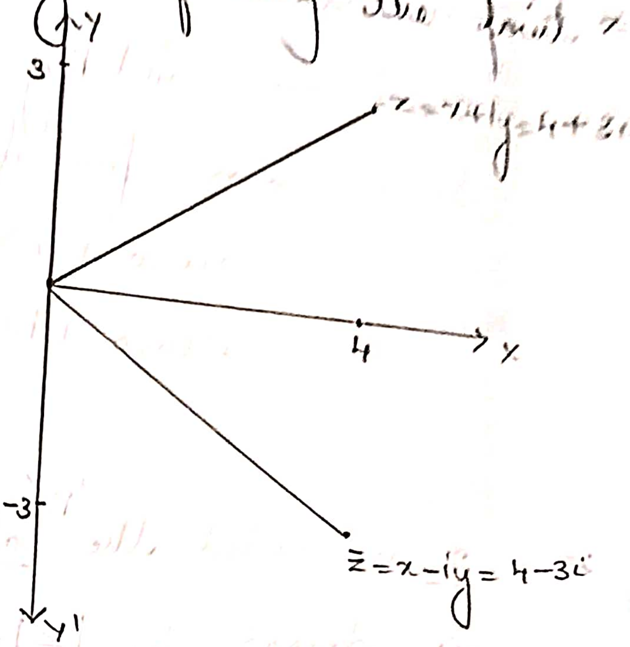


Subtraction of complex numbers

Complex Conjugate Numbers:

The complex conjugate $\bar{z}$ of a complex number $z = x + iy$ is defined by $\bar{z} = x - iy$.

It is obtained geometrically by reflecting the point z in the real axis.



ex: If $z = 4 + 3i$, then its conjugate is

$$\bar{z} = \overline{4+3i}$$
$$= 4 - 3i$$

Note: Let $z = x+iy$ and $\bar{z} = x - iy$.

- $z + \bar{z} = 2x = 2\,Re(z)$
- $z - \bar{z} = 2iy = 2i\,Im(z)$
- $z\bar{z} = x^2 + y^2 = |z|^2 = |\bar{z}|^2$
- $Re(z) = x = \frac{1}{2}(z + \bar{z})$
- $Im(z) = y = \frac{1}{2i}(z - \bar{z})$
- If z is real, $z = x$, then $\bar{z} = z$.
- $\overline{z_1 + z_2} = \bar{z}_1 + \bar{z}_2$
- $\overline{z_1 - z_2} = \bar{z}_1 - \bar{z}_2$
- $\overline{(z_1 z_2)} = \bar{z}_1 \cdot \bar{z}_2$
- $\overline{\left(\frac{z_1}{z_2}\right)} = \frac{\bar{z}_1}{\bar{z}_2}$.

$$z + \bar{z} = (x+iy) + \overline{(x+iy)}$$
$$= (x+iy) + (x-iy)$$
$$= 2x.$$

$$z - \bar{z} = (x+iy) - \overline{(x+iy)}$$
$$= (x+iy) - (x-iy)$$
$$= 2iy.$$

$$z\bar{z} = (x+iy)\overline{(x+iy)}$$
$$= (x+iy)(x-iy)$$
$$= x^2 - ixy + iyx - i^2y^2$$
$$= x^2 + y^2 \quad [\because i^2 = -1]$$

Polar form of Complex numbers

The complex number $z = x + iy$ can be represented by the point P whose cartesian coordinates are (x, y). If polar coordinates of the same point P are (r, θ), then

$\cos\theta = \frac{x}{r}$
$\sin\theta = \frac{y}{r}$

$x = r\cos\theta$ and $y = r\sin\theta$.

Hence, polar form of z is

$$z = r\cos\theta + ir\sin\theta$$
$$= r(\cos\theta + i\sin\theta)$$

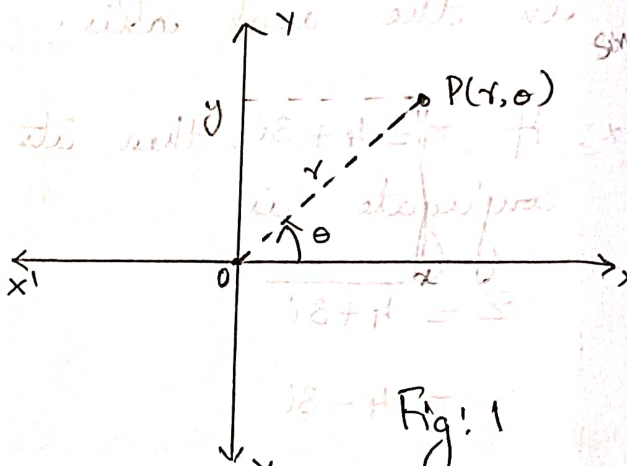


Fig: 1

- r is called the absolute value or modulus of z and is denoted by $|z|$. Hence,

$$|z| = r = \sqrt{x^2+y^2} = \sqrt{z\bar{z}}$$

Geometrically, $|z|$ is the distance of the point z from the origin. Similarly, $|z_1 - z_2|$ is the distance between z_1 and z_2.

- θ is called the argument of z and is denoted by $\arg z$.

Thus, $\theta = \arg z$ and $\tan\theta = \frac{y}{x}$.

Geometrically, θ is the directed angle from the positive X-axis to OP in Fig: 1.

Note that angles are measured in radians and positive in the counterclockwise direction.

- Hence, $|z| = r = \sqrt{x^2+y^2}$

$$\arg(z) = \theta = \tan^{-1}\left(\frac{y}{x}\right)$$

Note: The value of θ which satisfies both the equations $x = r\cos\theta$ and $y = r\sin\theta$, gives the argument of z.

- Argument θ has infinite number of values. The value of θ lying b/w $-\pi$ and π is called the principal value of argument denoted by $\operatorname{Arg} z$.

$\tan 30 = \tan 390$
$= \tan(360+30)$
$= \frac{1}{\sqrt{3}}$

$$\Rightarrow \quad -\pi < \operatorname{Arg} z \leq \pi.$$

- $|z_1 z_2| = |z_1| \cdot |z_2|$ and

$$\arg(z_1 z_2) = \arg(z_1) + \arg(z_2).$$

- Triangle inequality: $|z_1 + z_2| \leq |z_1| + |z_2|$.

In general, $|z_1 + z_2 + \dots + z_n| \leq |z_1| + |z_2| + \dots + |z_n|$

- $\left|\frac{z_1}{z_2}\right| = \frac{|z_1|}{|z_2|}$ and $\arg\left(\frac{z_1}{z_2}\right) = \arg z_1 - \arg z_2$.

<u>Exponential form of complex numbers:</u>

Let z be a complex number such that

$z = x+iy = r(\cos\theta + i\sin\theta)$.

where $x = r\cos\theta$, $y = r\sin\theta$. Then $r = \sqrt{x^2+y^2}$, $\theta = \tan^{-1}\left(\frac{y}{x}\right)$.

We know that, $e^{i\theta} = \cos\theta + i\sin\theta$.

Using polar form, $z = r(\cos\theta + i\sin\theta) = re^{i\theta}$.

This is called the exponential form or Euler's form of a complex number z.

Note: $e^{i\theta} = \cos\theta + i\sin\theta$

$e^{-i\theta} = \cos\theta - i\sin\theta$.

Hence: $\cos\theta = \dfrac{e^{i\theta} + e^{-i\theta}}{2}$

$\sin\theta = \dfrac{e^{i\theta} - e^{-i\theta}}{2i}$

$z = re^{i\theta}$

Integer powers of z: $z^n = r^n(\cos n\theta + i\sin n\theta)$

$z^n = (re^{i\theta})^n$

$= r^n e^{in\theta}$

$= r^n(\cos n\theta + i\sin n\theta)$

De Moivre's formula: $(\cos\theta + i\sin\theta)^n = \cos n\theta + i\sin n\theta$.

If $r = 1$

$z = e^{i\theta}$

$z^n = (e^{i\theta})^n$

$= \cos n\theta + i\sin n\theta$

Roots of a complex number:

A number ω is said to be an n^{th} root of a complex number z if $\omega^n = z$.

If we let $\omega = R(\cos\phi + i\sin\phi)$ and $z = r(\cos\theta + i\sin\theta)$

then

$\omega^n = z \Rightarrow [R(\cos\phi + i\sin\phi)]^n = r(\cos\theta + i\sin\theta)$

$\Rightarrow R^n(\cos n\phi + i\sin n\phi) = r(\cos\theta + i\sin\theta)$

$\Rightarrow R^n = r$ & $\cos n\phi + i\sin n\phi = \cos\theta + i\sin\theta$ —— (I)

From (I), $R^n = r \Rightarrow R = r^{1/n}$.

From (I), $\cos n\phi + i\sin n\phi = \cos\theta + i\sin\theta$.

$\Rightarrow \cos n\phi = \cos\theta$ and $\sin n\phi = \sin\theta$

(equating the real and imaginary parts)

$\Rightarrow n\phi = \theta + 2k\pi$, $k \in \mathbb{Z}$ (integer)

$\Rightarrow \phi = \frac{\theta}{n} + \frac{2k\pi}{n}$ where k is an integer.

For $k = 0, 1, 2, \ldots, n-1$, we get n distinct values for ω.
Further integers of k would give values already obtained.

Hence,

$$\omega = z^{1/n} \quad \text{(A)}$$

$$= r^{1/n}\left[\cos\left(\frac{\theta + 2k\pi}{n}\right) + i\sin\left(\frac{\theta+2k\pi}{n}\right)\right], \text{ where } k = 0, 1, 2, \ldots, n-1.$$

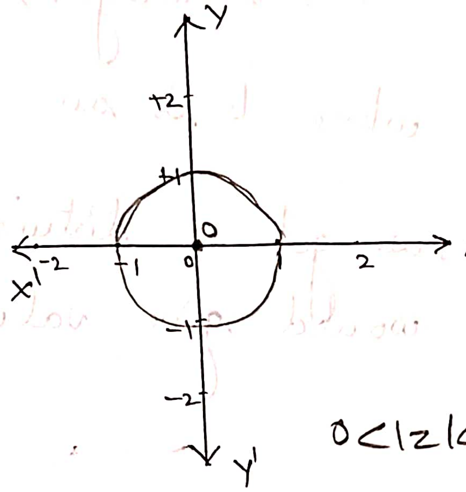
These n values lie on a circle of radius $r^{1/n}$ with center at the origin and constitute the vertices of a regular polygon of n sides. The value of $(z)^{1/n}$ obtained by taking the principal value of $\arg z$ and $k=0$ in the equation (A); is called the principal value of $\omega = z^{1/n}$.

Determine and sketch the points in the complex plane:

1. $0 < |z| < 1$.

$|z| < 1$ is an open disc centered at 0 and radius 1.

Here $0 < |z| < 1 \Rightarrow$ the point '0' is excluded from the region.



$0 < |z| < 1$.

2. $|\arg z| < \pi/4$.

$|x| \leq 2$

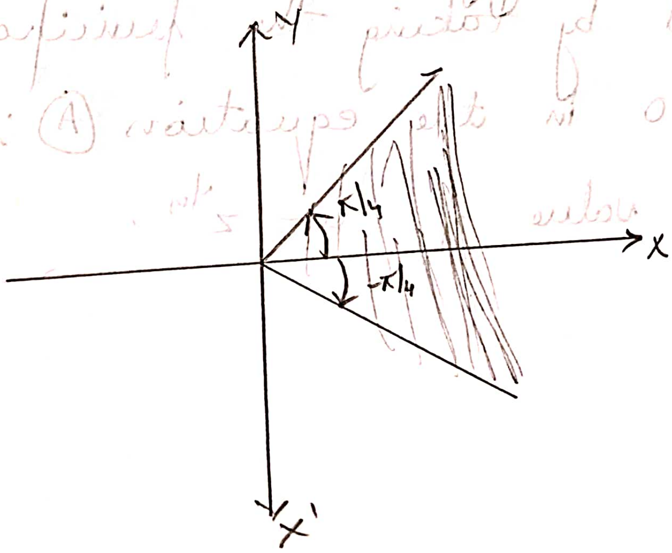
$\Rightarrow -2 \leq x \leq 2$

$|x| < 2$

$\Rightarrow -2 < x < 2$

$\Rightarrow -\pi/4 \leq \arg z \leq +\pi/4$.

The region consists of all points in z-plane with argument less than $\pi/4$ and greater than $-\pi/4$.



1. Find the n^{th} roots of unity or find solutions of $z^n = 1$.

$$z^n = 1 \Rightarrow z = 1^{1/n}.$$

Consider $z = 1$.

$\therefore r = |z| = 1$, $\text{Arg } z = 0$

$\therefore 1 = \cos 0 + i \sin 0$.

$$\therefore z = (1)^{1/n} = (\cos 0 + i \sin 0)^{1/n}$$

$$= \cos\left(\frac{0}{n} + \frac{2k\pi}{n}\right) + i \sin\left(\frac{0}{n} + \frac{2k\pi}{n}\right); \quad k = 0, 1, 2, \ldots, n. \quad \text{— (1)}$$

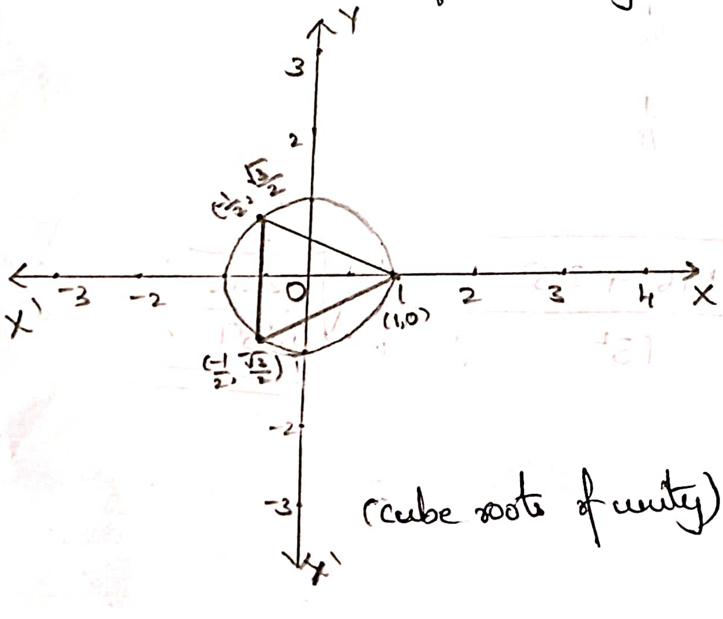
$$\Rightarrow z = \cos \frac{2k\pi}{n} + i \sin \frac{2k\pi}{n}, \quad k = 0, 1, 2, \ldots, n.$$

These n values are called n^{th} roots of unity.

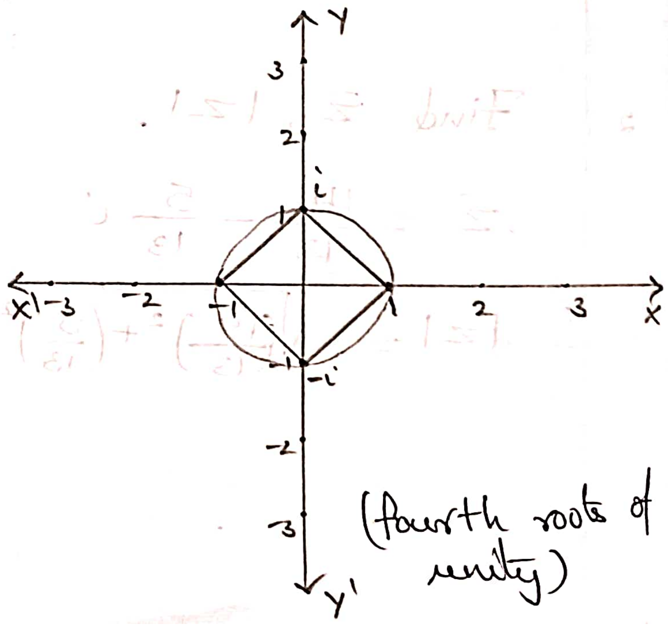
In particular, when $n = 3$, $1^{1/3} = 1, -\frac{1}{2} \pm \frac{1}{2}\sqrt{3}\,i$

When $n = 4$, $(1)^{1/4} = -1, \pm i$.

If ω denotes the value corresponding to $k = 1$ in (1), then the n^{th} root of unity are $1, \omega, \omega^2, \ldots, \omega^{n-1}$.



(cube roots of unity)



(fourth roots of unity)